Dynamical systems

- How does a system evolve in time?
- By a completely <u>deterministic</u> rule.
  - nothing random about it, although it may appear to be random, e.g. chaotic systems

<u>examples</u>:

$\dot{x} = f(x)$, $\quad x \in \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}^n$ (vector field)

$\dot{x} = \frac{dx}{dt}$

"autonomous", f does not depend on t

$\dot{x} = f(x,t)$

"non-autonomous"
$f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$

continuous time "flow"
discrete time "map"

$x_{n+1} = g(x_n)$

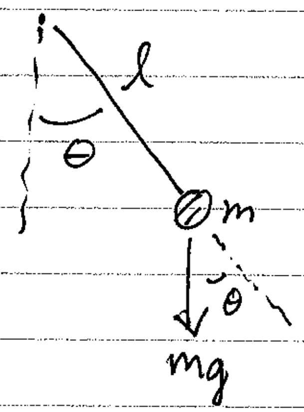
Note: in a typical differential eqns. class:

$$\left.\begin{aligned} \dot{x} &= f(x) \\ x(0) &= x_0 \end{aligned}\right\}$$ initial condition needed for unique soln. $x(t)$, $-T < t < T$, $T > 0$

Here we are interested in possible behavior as a function of x_0, and how it may change qualitatively with x_0.

We want to visualize/characterize solns in "phase space" or "state space", e.g. in all of $\mathbb{R}^n$ if $x \in \mathbb{R}^n$.

Example 1: idealized pendulum



Physics side-bar

$$\text{Torque} = \vec{r} \times \vec{F}$$
into board
$$= l\,mg\sin\theta\,(-\hat{k})$$
$$\text{Torque} = I\vec{\alpha}$$
$$= (ml^2)(\ddot{\theta}\,\hat{k})$$
$$ml^2\ddot{\theta} = -mlg\sin\theta$$
$$\ddot{\theta} = -\frac{g}{l}\sin\theta$$
$$= -\omega^2\sin\theta;\quad \omega^2 \equiv \frac{g}{l}$$

Let $\tau = \omega t$ = dimensionless time

$$\frac{d}{d\tau} = \frac{dt}{d\tau}\frac{d}{dt} = \frac{1}{\omega}\frac{d}{dt}$$

$$\frac{d^2\theta}{d\tau^2} = \frac{1}{\omega^2}\frac{d^2\theta}{dt^2} = \frac{1}{\omega^2}(-\omega^2\sin\theta) = -\sin\theta$$

$$\boxed{\frac{d^2\theta}{d\tau^2} = -\sin\theta}$$

solns. depend on

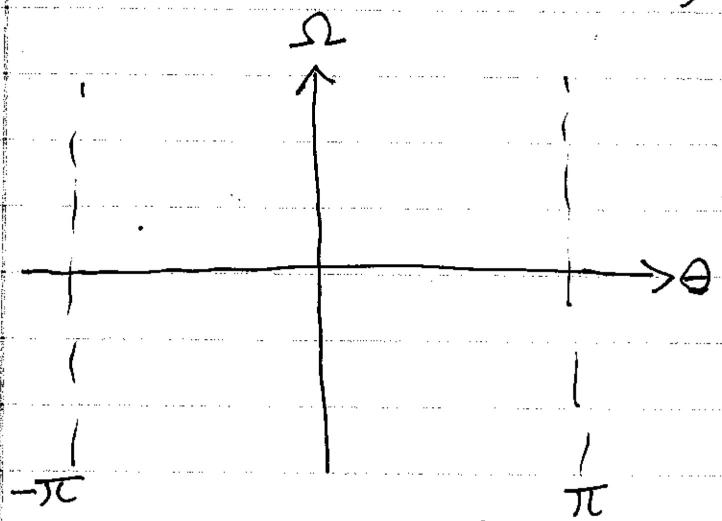
$$\underbrace{\Big(\underset{\text{initial angle}}{\theta(0)},\ \underset{\text{initial angular speed}}{\frac{d\theta}{d\tau}(0)}\Big)}_{\text{2 degrees of freedom}}$$

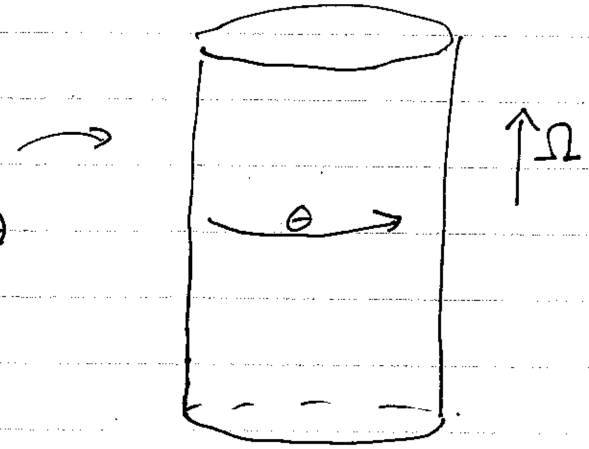
Can re-write as 2 first order eqns. using a standard trick:

let $\frac{d\theta}{d\tau} = \Omega$

then $\frac{d\Omega}{d\tau} = -\sin\theta$

"phase space" variables are (θ, Ω) which completely specify the state of the system at any given time t.





phase plane $= \mathbb{R}^2$

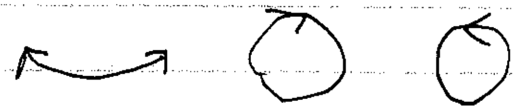
oops: $\theta \in [-\pi, \pi]$

phase space is a cylinder

$$\underset{\text{angle}}{S^1} \times \underset{\text{angular speed}}{\mathbb{R}}$$

Distinct types of solns.:

- 2 equilibria $\qquad (\theta,\Omega) = (0,0),\ (\pi,0)$
- 3 types of oscillations



physical space.
What do they look like in phase space?

Conservation of energy:

$$\underbrace{\frac{d\theta}{d\tau}\frac{d^2\theta}{d\tau^2}}_{\frac{1}{2}\frac{d}{d\tau}\left(\frac{d\theta}{d\tau}\right)^2} = \underbrace{-\sin\theta\,\frac{d\theta}{d\tau}}_{\frac{d}{d\tau}(\cos\theta)}$$

$$\frac{d}{dt}\left[\frac{1}{2}\Omega^2 - \cos\theta\right] = 0$$

kinetic energy $\qquad$ potential energy

$$\frac{1}{2}\Omega^2 - \cos\theta = E_0 = \text{(initial) energy} = \text{constant in time for solns } (\theta(\tau),\Omega(\tau))$$

solns in phase space are level curves of

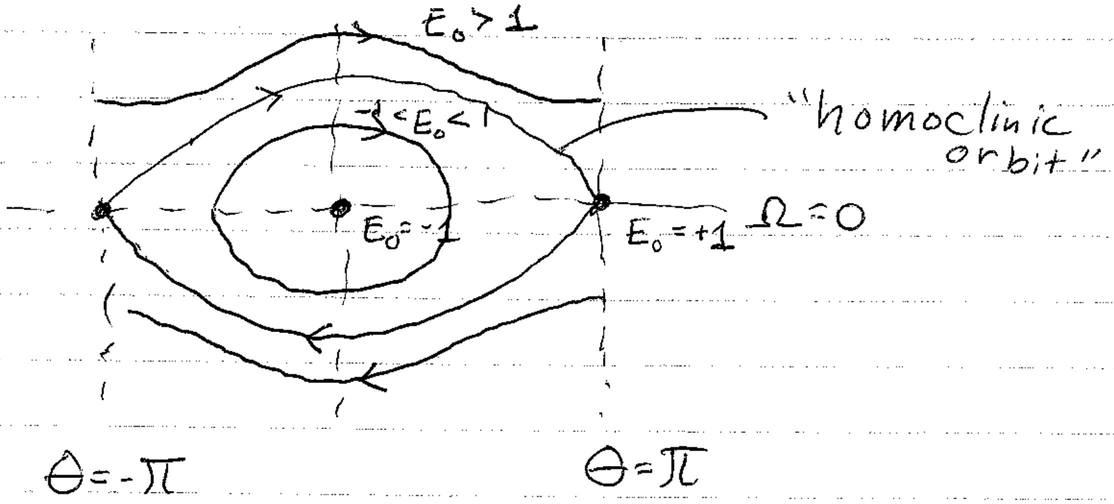
$$\underbrace{\frac{1}{2}\Omega^2}_{\geq 0} - \underbrace{\cos\theta}_{\in[-1,1]} \geq -1$$

$E_0 = -1 \Rightarrow \Omega = 0,\ \theta = 0$

$-1 < E_0 < 1 \Rightarrow \theta$ restristed

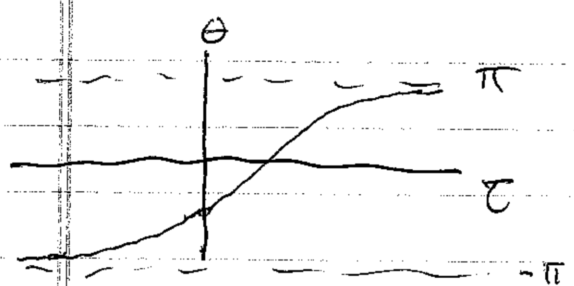
$E_0 = +1$

$E_0 > +1 \Rightarrow \Omega^2 > 0$ $\quad \Omega > 0$, $\Omega < 0$



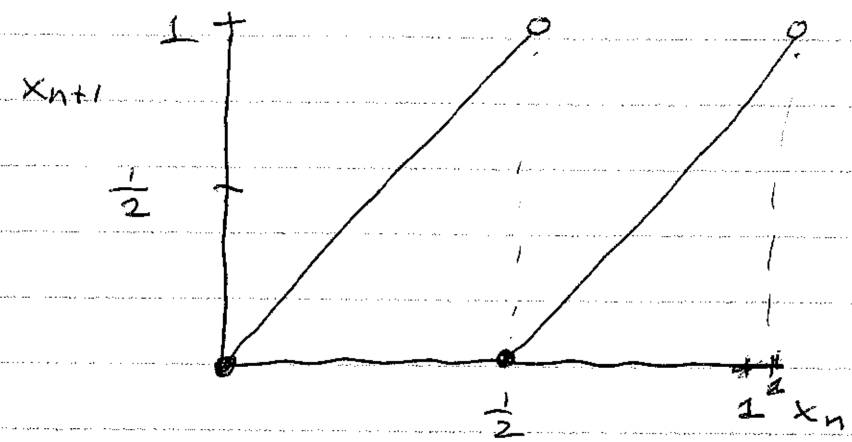
homoclinic orbit:

$$\left.\begin{aligned} \lim_{\tau \to +\infty} \theta(\tau) &= \pi \\ \lim_{\tau \to -\infty} \theta(\tau) &= -\pi \end{aligned}\right\} \text{ same equilibrium pt. of the cylinder}$$



Example 2: a piecewise linear map

$$X_{n+1} = 2x_n \pmod 1 = T(x_n), \quad x \in [0,1)$$



Claims about $T(x_n)$:
It has

properties of chaotic systems
(i) "sensitive dependence on initial conditions"
(ii) infinitely many periodic & aperiodic points
(iii) a dense orbit

(i) Sensitive dependence on initial conditions

orbit 1 $\quad x_0, x_1, x_2, \ldots \quad T(x_n) = x_{n+1} \quad x_0 \neq y_0$
orbit 2 $\quad y_0, y_1, y_2, \ldots \quad T(y_n) = y_{n+1}$

let $\epsilon = |y_0 - x_0|$ and $\delta_n = |y_n - x_n|$
δ_n grows to the bound of 1 for any ϵ.

(ii) periodic pts

if (minimal) period is N then

$x_1, x_2, x_3, x_4, \ldots x_N, x_1, x_2 \cdots$

ie. $x_{N+k} = x_k$

aperiodic $\Rightarrow$ doesn't repeat

(iii) a dense orbit

$x_0, x_1, x_2, \ldots$

$\exists x_0$ s.t. orbit comes arbitrarily close to each pt. in the interval $[0,1)$

Given any $\epsilon > 0$ and any pt. $p \in [0,1)$ then orbit $\{x_0, x_1, \ldots\}$ is dense if there is some finite N s.t. $|x_N - p| < \epsilon$.

~~Trick~~

Trick for showing all these properties:
write x_0 in binary as follows

$$\hat{x}_0 = 0.\underbrace{a_1 a_2 a_3 a_4 \ldots}_{a_k \in \{0,1\}} \qquad \text{in binary}$$

$a_0 = 0$

in decimal form:

$$x_0 = \sum_{j=1}^{\infty} a_j \left(\tfrac{1}{2}\right)^j$$

$$x_1 = 2x_0 \pmod 1$$

$$= \sum_{j=1}^{\infty} a_j \left(\tfrac{1}{2}\right)^{j-1} \pmod 1$$

$$= a_1 + \sum_{j=2}^{\infty} a_j \left(\tfrac{1}{2}\right)^{j-1} \pmod 1$$

$$= \cancel{\sum_{j=1}^{\infty}} \cancel{a_1} + \sum_{k=1}^{\infty} a_{k+1} \left(\tfrac{1}{2}\right)^k \qquad \begin{array}{l} j-1=k \\ j=k+1 \end{array}$$

$$\hat{x}_1 = 0.a_2 a_3 a_4 \ldots$$

$$\hat{x}_2 = 0.a_3 a_4 a_5 \ldots$$

action of map is just to shift decimal by 1.
periodic pts are repeating & aperiodic not.

Sensitive dependence on initial conditions

$\hat{x}_0 = 0, a_1 a_2 a_3 \cdots a_N \, a_{N+1} \, a_{N+2} \cdots$

$\hat{y}_0 = 0, a_1 a_2 a_3 \cdots a_N \, b_1 \, b_2 \cdots$

N set by ϵ

$$b_j = \begin{cases} 0 & \text{if } a_{N+j} = 1 \\ 1 & \text{if } a_{N+j} = 0 \end{cases}$$

after N iterations of map, the orbits are as far apart as possible

iii dense orbit by construction

Construct all sequences of length k:

k=1	0, 1	2
k=2	00, 01, 10, 11	4
k=3	000, 001 ..., ... 111	8

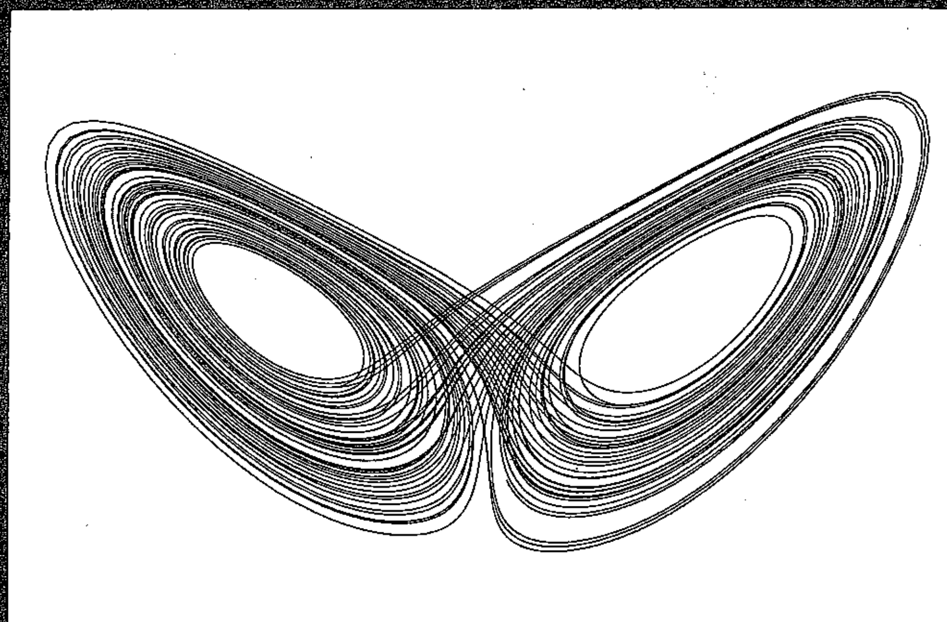
and concatenate

$x_0 = .01000110110000 01 \cdots$

CAMBRIDGE TEXTS IN APPLIED MATHEMATICS

Stability, instability and chaos:

an introduction to the theory of nonlinear differential equations



PAUL GLENDINNING

$\frac{8}{3}z + xy.$ (11.1c)

ions are shown in Figure 11.1, where
ions have some geometric structure,
o a simple periodic orbit. We shall
of systems like the Lorenz equations

Dynamical Systems, was published
icle about a particular class of maps
igorous mathematical results about
These two papers show up some of
lking about chaos. For the applied
n is not chaotic unless it has what is
mplicated behaviour must be visible
e interested in the topological struc-
a system chaotic even if the chaotic
to know that it is there. Hence the
and co-workers is not especially in-
chaotic behaviour is not attracting,
such as the Lorenz equations are so
to prove any rigorous results about
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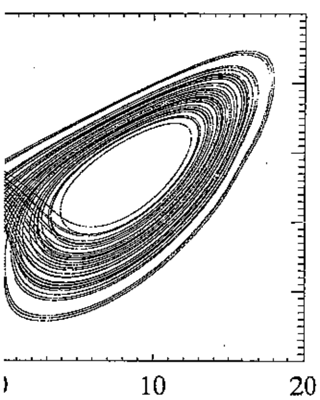
os. In order to prove results we shall
e it is remarkably difficult to prove
in all but the most artificial of ex-



orenz attractor.

amples. None the less, we shall remain aware of the dichotomy between the study of physical examples and the rigours of theory and wherever possible give results about attractors.

This chapter begins with some basic definitions and examples, then goes on to develop the theory of some particularly simple maps which have complicated (or chaotic) behaviour.

11.1 Characterizing chaos

The first step towards developing any theory is to try to establish the most important features of the problem. As remarked above this is not completely clear for chaotic systems and depends crucially on whether one takes a topological point of view or chooses to ignore anything which is not an attractor (see, for example, Exercise 1 at the end of this chapter). An example may help to crystallize ideas.

Example 11.1

Consider the simple map $T : [0,1) \to [0,1)$ defined by

$$T(x) = 2x \text{ (mod 1)}.$$

Seen as a map from the interval to itself this map has a discontinuity at $x = \frac{1}{2}$ which can cause problems (see Fig. 11.2). However, this example

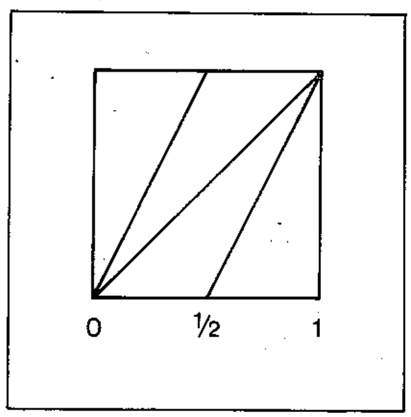


Fig. 11.2 The map $T(x) = 2x(\text{mod } 1)$.

can be thought of as being derived from the complex map

$$z_{n+1} = z_n^2$$

by noting that if $|z_n| = 1$ then $|z_{n+1}| = 1$, so setting $z_n = e^{2\pi i x_n}$ we find that $x_{n+1} = 2x_n$ and since x is an angle it can be thought of as taking values between 0 and 1. Hence although the map is a discontinuous map of the interval it represents a continuous map of the circle. Any point $x \in [0,1]$ can be written as a binary expansion in the form

$$x = \sum_{i=1}^{\infty} \frac{a_i}{2^i}$$

where $a_i \in \{0,1\}$, and so

$$2x = a_1 + \sum_{i=1}^{\infty} \frac{a_{i+1}}{2^i}.$$

Hence, if we identify 1 with 0,

$$T(x) = \sum_{i=1}^{\infty} \frac{a_{i+1}}{2^i}.$$

The effect of T is now clear: if x has a binary expansion $[a_1, a_2, a_3, ...]$ then $T(x)$ has binary expansion $[a_2, a_3, ...]$, $T^2(x)$ has binary expansion $[a_3, ...]$ and so on. It is now possible to deduce that

i) T has 2^n periodic points of period n (but not necessarily least period n), since there are 2^n different sequences of 0s and 1s of period n;
ii) if $x \neq y$ then $|T^n(x) - T^n(y)|$ grows initially like $2^n|x-y|$, since the slope of the map is 2 everywhere;
iii) given any neighbourhood J of x, there exists a subinterval K of J and $n > 0$ such that $T^n(J) = (0,1)$ and $T^n|J$ is a homeomorphism, this is proved below;
iv) periodic points are dense in [0,1], this follows from (iii): given x and a neighbourhood J, define $K = [a,b]$ and n as in (iii), then $T^n(a_+) = 0 \leq a$ and $T^n(b_-) = 1 \geq b$, now since T^n is continuous on K this implies that T^n has a fixed point in K (by the Intermediate Value Theorem) and hence T has a periodic point in K. Alternatively, simply note that the periodic points are points with periodic binary expansions. Such points are dense in $[0,1]$;
v) T has a dense orbit; this is by construction: let A_{mk} be a list of the possible sequences of 0s and 1s of length m, so $1 \leq k \leq 2^m$,

and let x be the point with binary

$$[A_{11}, A_{12}, A_{21}, A_{22}, A_{23},$$

then $T^n(x)$ passes arbitrarily close

The only property that we have nc uses techniques which will be useful in will dignify the statement by calling i is an ambiguity in the binary expansio $[1,0,0,0,\ldots] = [0,1,1,1,\ldots]$. This refle fined on the circle on which the points not worry about this here.

(11.1) LEMMA

Consider the map $T : [0,1) \to [0,1)$ defin the convention that $T(\frac{1}{2}) = 0$. Then fc there exists an open neighbourhood, J, that $T^n(J) = (0,1)$ and $T^n|J$ is a homeo

Proof: Suppose that x has binary expans a preimage of $\frac{1}{2}$ (so the binary expansio string of 0s or 1s). All points close to x agree with the binary expansion of x up N sufficiently large we can arrange for all $[a_1, \ldots, a_N, b_1, b_2, \ldots]$ to lie inside an ϵ n not a preimage of $\frac{1}{2}$ we can choose $M >$ let y be the point with binary expansion

$$[a_1, \ldots, a_M, 0, 0, 0,$$

It should be obvious that $y < x$ since

$$x - y = \sum_{M+1}^{\infty} \frac{a_i}{2^i}$$

and, by assumption, $a_i \neq 0$ for some $i >$

Similarly, let z be the point with binar

$$[a_1, \ldots, a_M, 1, 0, 0,$$

and note that since $a_{M+1} = 0$, $z > x$. (general that if

$$x = [a_1, \ldots, a_k, 0, a_{k+1}, \ldots] \text{ and } z =$$

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$$z_{n+1} = z_n^2$$

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$$x = \sum_{i=1}^{\infty} \frac{a_i}{2^i}$$

$$a_1 + \sum_{i=1}^{\infty} \frac{a_{i+1}}{2^i}.$$

$$) = \sum_{i=1}^{\infty} \frac{a_{i+1}}{2^i}.$$

x has a binary expansion $[a_1, a_2, a_3, ...]$
$[a_2, a_3, ...]$, $T^2(x)$ has binary expansion
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and let x be the point with binary expansion

$$[A_{11}, A_{12}, A_{21}, A_{22}, A_{23}, A_{24}, A_{31}, A_{32}, ...]$$

then $T^n(x)$ passes arbitrarily close to every point in (0,1).

The only property that we have not proved is (iii). The proof uses techniques which will be useful in a more general context, so we will dignify the statement by calling it a lemma. Note that there is an ambiguity in the binary expansion of some points, for example $[1, 0, 0, 0, \ldots] = [0, 1, 1, 1, \ldots]$. This reflects the fact that T is really defined on the circle on which the points 0 and 1 are identified. We will not worry about this here.

(11.1) LEMMA

Consider the map $T : [0,1) \to [0,1)$ defined by $T(x) = 2x \pmod 1$, with the convention that $T(\frac{1}{2}) = 0$. Then for all $x \in (0,1)$ and all $\epsilon > 0$ there exists an open neighbourhood, J, of x of length less than ϵ such that $T^n(J) = (0,1)$ and $T^n|J$ is a homeomorphism.

Proof: Suppose that x has binary expansion $[a_1, a_2, \ldots]$ and that x is not a preimage of $\frac{1}{2}$ (so the binary expansion does not end with an infinite string of 0s or 1s). All points close to x have binary expansions which agree with the binary expansion of x up to some stage, N. By choosing N sufficiently large we can arrange for all points with binary expansions $[a_1, \ldots, a_N, b_1, b_2, \ldots]$ to lie inside an ϵ neighbourhood of x. Since x is not a preimage of $\frac{1}{2}$ we can choose $M > N$ such that $a_{M+1} = 0$. Now let y be the point with binary expansion

$$[a_1, \ldots, a_M, 0, 0, 0, 0, 0, \ldots].$$

It should be obvious that $y < x$ since

$$x - y = \sum_{M+1}^{\infty} \frac{a_i}{2^i} > 0$$

and, by assumption, $a_i \neq 0$ for some $i > M$.

Similarly, let z be the point with binary expansion

$$[a_1, \ldots, a_M, 1, 0, 0, 0, 0, \ldots]$$

and note that since $a_{M+1} = 0$, $z > x$. (You should prove this: show in general that if

$$x = [a_1, \ldots, a_k, 0, a_{k+1}, \ldots] \text{ and } z = [a_1, \ldots, a_k, 1, b_{k+1}, \ldots],$$

i.e. if the binary expansion of x and z agree in the first k terms and then x has a 0 whilst z has a 1, then $x < z$ regardless of the subsequent expansion.)

We now claim that $T^m(y,z) = (0, \frac{1}{2})$ and that $T^m|(y,z)$ is a homeomorphism, which proves the lemma since $T(0, \frac{1}{2}) = (0,1)$ and T is a homeomorphism on $(0, \frac{1}{2})$. The claim is proved in two parts.

i) First we show that if the binary expansions of two points agree for the first k terms then T^k is a homeomorphism when restricted to the interval between these points.
ii) The definition of the action of T on the binary expansion of points then allows us to conclude that $T^M(y) = 0$ and $T^M(z) = \frac{1}{2}$. This, together with the fact that T^M is a homeomorphism on (y,z), completes the proof.

The proof of (i) is the most difficult; (ii) is obvious. Suppose that the binary expansions of y and z agree for the first k terms. This implies that the binary expansions of all points between them also agree in the first k terms. (If this is not obvious to you, prove it.) If the first term of the binary expansion of a point is 0 then that point lies in $[0, \frac{1}{2}]$, whilst if it starts with 1 then the point lies in $[\frac{1}{2}, 1]$. Consider $T(y,z)$. Since all points between y and z start with the same symbol, $T|(y,z)$ is a homeomorphism and (if $k > 1$) all points in the image of (y,z) start with the same symbol and $T(y,z) = (T(y), T(z))$ since T is monotonic and increasing on (y,z). Repeating this argument k times we find that $T^k|(y,z)$ is a homeomorphism onto its image, which is $(T^k(y), T^k(z))$.

Now, for the choice $k = M$ and y and z defined in the beginning of the proof, we see that $T^M|(y,z)$ is a homeomorphism onto its image, which is $(0, \frac{1}{2})$ by construction. Hence $T^{M+1}(y,z) = (0,1)$.

This example has picked out various properties that one might wish to use as definitions of chaos. It is now only a question of choosing some minimal list to define what we will mean by chaotic. The definition of chaos below is based upon a generalization of the previous example, which is called a horseshoe.

(11.2) DEFINITION

Let f be a continuous map of the interval, I. The map f has a horseshoe iff there exists J contained in I and disjoint open subintervals K_1 and K_2 of J such that $f(K_i) = J$ for $i = 1, 2$.

Before describing the consequences of give one further definition.

(11.3) DEFINITION

A continuous map $f : I \to I$ is chaotic iff $n \geq 1$.

Although this definition of chaos may se ample we will see later that it is, in fact, eq give the impression of being considerably r for maps of the interval (and diffeomorphis shoe is not simply an example of chaotic be chaotic behaviour. The consequences of a r almost precisely the same as the results obt

(11.4) LEMMA

Suppose that the continuous map $f : I \to I$ h

i) f^n has at least 2^n fixed points;
ii) f has periodic points of every period;
iii) f has an uncountable number of aperio

The proof of this lemma is the same as th statements for Example 11.1. However, f may and K_2 and may have stable periodic orbits v the more exciting parts of the proof. We leave As remarked above there are many other c completeness we shall give three other charac give an idea of how they relate to the definitior

(11.5) DEFINITION

Let f be a continuous map of the interval or a s A set S is (n, ϵ)-separated iff for all x and y in S $0 \leq k \leq n$, such that $|f^k(x) - f^k(y)| > \epsilon$.

Thus the cardinality of (n, ϵ)-separated sets gi ber of distinguishable orbits of f at resolution